

Problem 11.20

Investment X of 100,000 is invested at a nominal rate of interest j convertible semiannually. After four years, it accumulates to 214,358.88. Investment Y of 100,000 is invested at a nominal rate of discount k convertible quarterly. After two years, it accumulates to 232,305.73.

Investment Z of 100,000 is invested at an annual effective rate of interest equal to j in year one and an annual effective rate of discount equal to k in year two. Calculate the value of investment Z at the end of two years.

Solution

For investment X , there are eight semiannual compounding periods in four years. Therefore,

$$\begin{aligned} 100,000 \left(1 + \frac{j}{2}\right)^8 &= 214,358.88, \\ j &= 2 \left[\left(\frac{214,358.88}{100,000} \right)^{1/8} - 1 \right] \\ &\approx 0.20. \end{aligned}$$

Thus, $j = 20\%$.

For investment Y , there are eight quarterly conversion periods in two years. Hence,

$$\begin{aligned} 100,000 \left(1 - \frac{k}{4}\right)^{-8} &= 232,305.73, \\ k &= 4 \left[1 - \left(\frac{232,305.73}{100,000} \right)^{-1/8} \right] \\ &\approx 0.40. \end{aligned}$$

Thus, $k = 40\%$.

In the first year, investment Z earns an annual effective interest rate of 20%. In the second year, it earns an annual effective discount rate of 40%, whose accumulation factor is $(1 - 0.40)^{-1}$. Therefore,

$$\begin{aligned} Z_2 &= 100,000(1 + 0.20)(1 - 0.40)^{-1} \\ &= 100,000(1.20) \left(\frac{1}{0.60} \right) \\ &= 200,000. \end{aligned}$$

Thus, the value of investment Z at the end of two years is

$$\boxed{200,000}.$$